Swoyam Suwal

976765992 | swoyamsuwal@proton.me | LinkedIn | GitHub

Summary

Software Engineer focused on system and AI. Experience building REST APIs, authentication systems, and (RAG) Retrieval-Augmented Generation workflows using Django REST Framework, Laravel, pgvector and LLM integrations. Built full-stack and backend systems involving semantic search, JWT authentication, role-based access control, MinIO object storage and email automation. Seeking AI Engineering, Backend Development, or Software Engineering.

Education

Herald College Kathmandu, Nepal
BSc (Hons) in Computing – awarded by University of Wolverhampton *2023 – Present*

Little Angels' College Kathmandu, Nepal
Class XII *2023*

Little Angels' School Kathmandu, Nepal
Class X *2021*

Experience

Trainee – PHP Developer Oct 2024 – Dec 2024
Outlines Research & Development Pvt. Ltd. *Kathmandu, Nepal*

- Developed and optimized CRUD modules and RESTful APIs in Laravel to support reliable frontend–backend data exchange.
- Implemented authentication-related features including session handling and email-based password reset workflows.
- Built admin dashboard modules for monitoring users, sales, and event participation through dynamic reporting views.
- Contributed backend development across APF Nepal, Event Management App, and Saathi projects.

Intern – PHP Developer Jul 2024 – Oct 2024
Outlines Research & Development Pvt. Ltd. *Kathmandu, Nepal*

- Built backend features using Laravel, MySQL, and PostgreSQL, including schema design and API development.
- Created and tested RESTful APIs with Postman, validating request/response handling and edge cases.
- Improved input validation, security checks, and authentication flows across multiple application modules.
- Contributed to Siddhartha Cable Car, APF Nepal, and NRA Nepal projects.

Projects

Chronova – AI-Powered Office Assistant | *Django, Django REST Framework, LangChain, pgvector, MinIO, Ollama, Next.js* 2025

- Engineered a multi-module AI office assistant platform with Django REST Framework and Next.js, combining document management, task tracking, email automation, campaign management, and Google Calendar integration in a unified workspace.
- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain, recursive text splitting, Ollama embeddings and pgvector cosine similarity search to support semantic question answering across PDF, DOCX, PPTX, and text documents.
- Architected secure authentication and authorization using OTP-based login, session authentication, SimpleJWT, and a custom RBAC system with per-feature permissions for view, create, update, delete, and execute actions.
- Integrated a locally hosted LLaMA 3.2 model via Ollama for document Q&A, AI email drafting, and natural-language calendar actions backed by Google Calendar OAuth2 workflows.
- Implemented company-scoped file storage with MinIO/S3-compatible object storage, embedded document chunking, and isolated multi-tenant access control for users, files, tasks, campaigns, and roles.

Vadama – Room Rental | *Laravel, PHP, MySQL, Blade, Bootstrap* 2024

- Built a full-stack e-commerce platform with Laravel following MVC architecture, using Eloquent ORM for product catalog, banner, and customer account management.
- Implemented CRUD operations for product and banner management alongside role-protected admin functionality and responsive user-facing pages.
- Used Laravel's built-in authentication, routing, and middleware to build secure, maintainable backend services.

Certifications

Prompt Design in Vertex AI – Skill Badge Aug 2026
Google

Python Certification Jun 2023 – Jul 2023
Basic Python Course – IT Training Nepal

Technical Skills

Languages: Python, PHP, JavaScript, SQL
Frameworks & Libraries: Django, Django REST Framework, Laravel, Next.js, LangChain, CodeIgniter, Fast-API
Databases: PostgreSQL, MySQL, pgvector
Authentication & Security: JWT, SimpleJWT, Session Authentication, OTP Verification, RBAC, OAuth2
Developer Tools: Git, Docker, Postman, Linux, MinIO
AI / ML: Retrieval-Augmented Generation (RAG), Vector Embeddings, Semantic Search, LLMs, Ollama, Prompt Engineering
Concepts: RESTful APIs, Backend Development, Multi-tenant Systems, Object Storage, Email Automation, MVC Architecture